

BURLINGTON INTL, VT, USAWMO: **726170**Lat: **44.468N**Lon: **73.150W**Elev: **101**StdP: **100.12**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **14742****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-18.7	-27.0	0.3	-21.0	-24.2	0.4	-17.8	11.8	0.6	10.9	-1.9	2.8	80	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.4	21.8	29.9	21.0	28.3	20.3	23.6	28.8	22.6	27.5	21.7	26.4	4.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.9	16.7	26.1	20.9	15.8	25.3	20.0	14.9	24.5	70.7	28.9	66.9	27.2	63.7	26.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	10.1	8.8	8.0	-25.5	34.3	3.5	1.8	-28.0	35.6	-30.0	36.7	-32.0	37.7	-34.5	39.0	
(5)				-25.8	25.3	3.4	1.1	-28.2	26.1	-30.2	26.7	-32.1	27.3	-34.6	28.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-6.6	-5.3	-0.3	7.0	14.2	19.3	21.9	21.0	16.8	9.9	3.6	-2.5	(6)
(7)		DBStd	11.16	7.25	6.26	6.00	4.90	4.41	3.75	3.01	3.10	4.17	4.62	5.11	5.69	(7)
(8)		HDD10.0	2042	517	427	323	115	11	0	0	0	2	61	200	388	(8)
(9)		HDD18.3	3969	774	661	577	340	143	32	4	11	78	262	442	646	(9)
(10)		CDD10.0	1433	1	0	4	27	141	278	371	340	206	58	8	1	(10)
(11)		CDD18.3	318	0	0	0	2	14	61	117	92	31	2	0	0	(11)
(12)		CDH23.3	2619	0	0	3	23	149	526	974	693	239	11	0	0	(12)
(13)		CDH26.7	741	0	0	0	4	37	158	298	183	61	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.9	3.7	3.8	3.8	3.5	3.1	2.9	2.9	3.1	3.4	3.8	3.7	(14)
(15)	Precipitation	PrecAvg	881	49	40	55	71	81	88	97	100	90	76	76	58	(15)
(16)		PrecMax	1281	131	137	105	166	160	220	236	200	261	159	174	151	(16)
(17)		PrecMin	591	11	5	10	18	7	21	20	30	36	13	24	9	(17)
(18)		PrecStd	147	30	22	20	31	36	41	40	39	44	36	29	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	12.2	20.6	26.4	30.6	33.3	34.0	32.7	31.4	25.0	19.2	13.3	(19)
(20)			MCWB	11.7	8.3	13.6	16.3	19.5	22.6	23.8	22.7	22.0	17.6	14.5	10.2	(20)
(21)		2%	DB	8.7	8.2	14.4	22.1	27.1	30.5	31.4	30.4	28.3	22.0	16.4	10.1	(21)
(22)			MCWB	6.5	4.8	8.8	13.3	17.4	20.9	22.2	21.5	20.3	16.2	12.4	7.6	(22)
(23)		5%	DB	5.5	5.9	10.9	18.7	24.7	28.4	29.8	28.7	26.1	19.7	13.8	7.7	(23)
(24)			MCWB	3.4	3.0	6.5	11.1	16.1	19.9	21.4	20.7	19.5	14.6	10.0	5.3	(24)
(25)		10%	DB	3.0	3.7	8.3	15.5	22.4	26.3	28.1	27.1	23.9	17.3	11.5	5.2	(25)
(26)			MCWB	1.1	1.4	4.7	9.2	15.2	19.0	20.6	20.1	18.4	13.2	8.2	3.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	9.0	14.4	17.8	21.2	24.1	25.2	24.1	23.2	19.4	15.5	10.8	(27)
(28)			MCDWB	13.7	11.3	19.3	24.5	27.4	30.0	31.4	29.7	28.5	22.5	18.2	12.7	(28)
(29)		2%	WB	6.7	5.4	9.9	14.6	19.5	22.7	23.9	22.9	21.7	17.3	13.0	8.0	(29)
(30)			MCDWB	8.2	7.9	13.5	20.3	24.9	28.0	29.2	28.0	26.3	21.2	16.2	9.9	(30)
(31)		5%	WB	3.6	3.2	7.0	12.0	17.9	21.5	22.8	22.0	20.4	15.2	10.6	5.6	(31)
(32)			MCDWB	5.3	5.4	10.2	17.0	22.8	26.2	27.4	26.5	24.7	18.7	13.2	7.4	(32)
(33)		10%	WB	1.2	1.6	4.9	10.0	16.3	20.2	21.8	21.2	19.1	13.6	8.4	3.2	(33)
(34)			MCDWB	2.9	3.5	8.2	14.9	21.1	24.5	26.6	25.4	23.2	16.7	11.1	5.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	8.9	9.1	10.4	11.1	10.6	10.5	10.5	8.9	7.5	7.1	(35)	
(36)			MCDBR	9.2	9.8	13.1	15.9	14.7	13.4	12.1	11.9	12.8	11.7	11.0	8.5	(36)
(37)			MCWBR	8.4	7.9	8.9	8.9	7.2	5.9	5.2	5.2	6.2	6.9	8.4	7.8	(37)
(38)		5% WB	MCDBR	9.0	9.4	12.0	13.5	11.9	10.9	10.1	10.0	10.5	10.6	9.9	8.4	(38)
(39)			MCWBR	8.5	8.0	9.0	8.7	6.9	5.7	5.1	5.0	5.6	7.1	8.5	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.289	0.329	0.352	0.380	0.434	0.443	0.418	0.370	0.336	0.310	0.280	(40)	
(41)		taud	2.346	2.339	2.371	2.405	2.380	2.231	2.214	2.277	2.399	2.499	2.496	2.425	(41)	
(42)		Ebn at Noon	833	902	907	911	891	839	826	834	851	841	806	809	(42)	
(43)		Edh at Noon	78	94	105	111	117	137	138	125	102	81	68	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.43	2.27	3.39	4.29	5.01	5.41	5.53	4.97	4.02	2.36	1.49	1.05	(44)	
(45)		RadStd	0.16	0.22	0.23	0.52	0.55	0.48	0.42	0.23	0.31	0.25	0.18	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	-139	+109	+54	
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+67	N/A	(47)